# Island Satiates without Change in Backgroundedness

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**Introduction**: Certain structural domains block syntactic movements from within (*island effect*, [1]). For example, complex-NPs in English, an island structure, block *wh*-movement from within (example (1), complex-NP bracketed).

(1) *Who did John discuss [the claim that Mary saw __ ]?

The unacceptability of island-violating sentences like (1) have been variably attributed to syntax [1-3], processing difficulty [4,5], and discourse constraints [6-8]. Under the discourse account, island effects are explained as a clash between the discourse backgrounded status of island domains and the foregrounding function of *wh*-movements. This study puts the discourse account of islands to test using the *island satiation effect*—the phenomenon whereby island violations appear weaker to speakers after repeated exposure [9]. If islandhood reduces entirely to backgroundedness as suggested by [7-8], the satiation effect should reflect a reduction in the backgroundedness of the island domain. This study examines this prediction.

**Method**: The experiment (N = 300, *Prolific*) consists of three phases: pre-exposure phase, exposure phase, and post-exposure phase. In the pre- and post-exposure phases, we use the negation task [7,10] to probe the backgroundedness of complex-NP island domains. Participants read a declarative context sentence containing a complex NP and then answer a question targeting that NP using a slider scale (left end: *definitely no*; right end: *definitely yes*; midpoint: *can't tell*). Each item appears in one of two polarity conditions: baseline or negated (example (2)). Following [7,10], we assume that backgrounded constituents are not affected by sentential negation, whereas at-issue constituents are. Thus, the difference in responses between baseline and negated conditions indexes the degree of backgroundedness of the island domain (larger differences=lower backgroundedness). If complex-NP backgroundedness decreases after exposure, we expect a significant context condition×phase interaction.

(2) <u>Context:</u>(Baseline) *John discussed the claim that the boy ate a sandwich.*
(Negated) *John didn't discuss the claim that the boy ate a sandwich.*
<u>*Question*</u>*: Was there a claim that the boy ate a sandwich?*

The exposure phase uses a sentence-acceptability rating task with a 2×2 factorial design manipulating structure (island vs. non-island) and dependency length (short vs. long). Example stimuli are shown in Table 1. Participants read a target sentence and rate its acceptability on a slider scale. Following [11], we operationalize the island effect as the interaction between structure and dependency length (the super-additive penalty observed when a long dependency occurs inside a complex-NP island). The exposure phase consists of four unannounced blocks, each containing four critical sentences (one per condition). If the island effect satiates, we expect a significant three-way interaction among block, structure, and length, such that the super-additive structure × length interaction decreases across blocks.

**Results and Discussion**: Acceptability ratings from the exposure phase z-scored by participant (Figure 1). We fitted an LMER model predicting acceptability z-scores with structure, length, block order, and their interactions. There is a significant structure×length interaction ($\beta=0.44$, $p<0.001$), confirming a complex-NP island effect. There is also a significant three-way interaction ($\beta=-0.021$, $p<0.001$) where the island penalty decreased over blocks, indicating satiation. Negation task responses from the pre- and post-exposure phases are shown in Figure 2. We fitted an LMER model predicting the negation task scores with polarity, task position, and their interaction. There is a significant polarity effect ($\beta=0.037$, $p<0.001$), but no effect of task position ($p=0.99$) and no interaction ($p=0.47$). Thus, backgroundedness of the complex-NP domain did **not** change across phases, even though the island effect satiated.

**Conclusion:** Our results show a clear dissociation between island satiation and change in discourse backgroundedness. While the complex-NP island effect weakened with repeated exposure, the negation-task responses show no corresponding shift in the backgroundedness of

the island domain. This suggests that island satiation cannot be attributed to changes in discourse status, challenging a core prediction of the discourse-based account of islands.

Table 1. Example stimuli sentences for the exposure phase

| | Non-island | Island |
|---|---|---|
| Short | Who claimed that the boy ate a sandwich? | Who discussed the claim that the boy ate a sandwich? |
| Long | What did John claim that the boy ate? | What did John discuss the claim that the boy ate? |

Figure 1. Acceptability judgment results

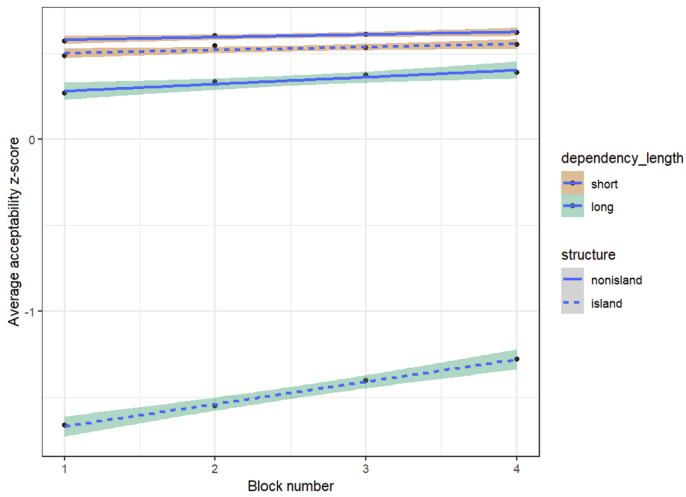
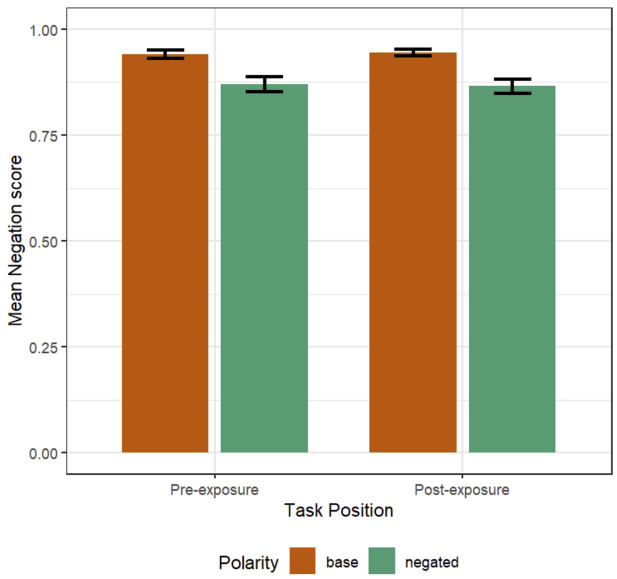



Figure 2. Negation task results





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